Data Assimilation
- A10. Adaptive Inflations -

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DA Lectures A (Basic Course)

- (1) Introduction and NWP
- (2) Deterministic Chaos and Lorenz-96 model
- (3) A toy model and Bayesian estimation
- (4) Kalman Filter (KF)
- (5) 3D Variational Method (3DVAR)
- (6) Ensemble Kalman Filter (PO method)
- (7) Serial Ens. Square Root Filter (Serial EnSRF)
- (8) Local Ens. Transform Kalman Filter (LETKF)
- (9) Innovation Statistics
- (10) Adaptive Inflations
- (11) 4D Variational Method (4DVAR)

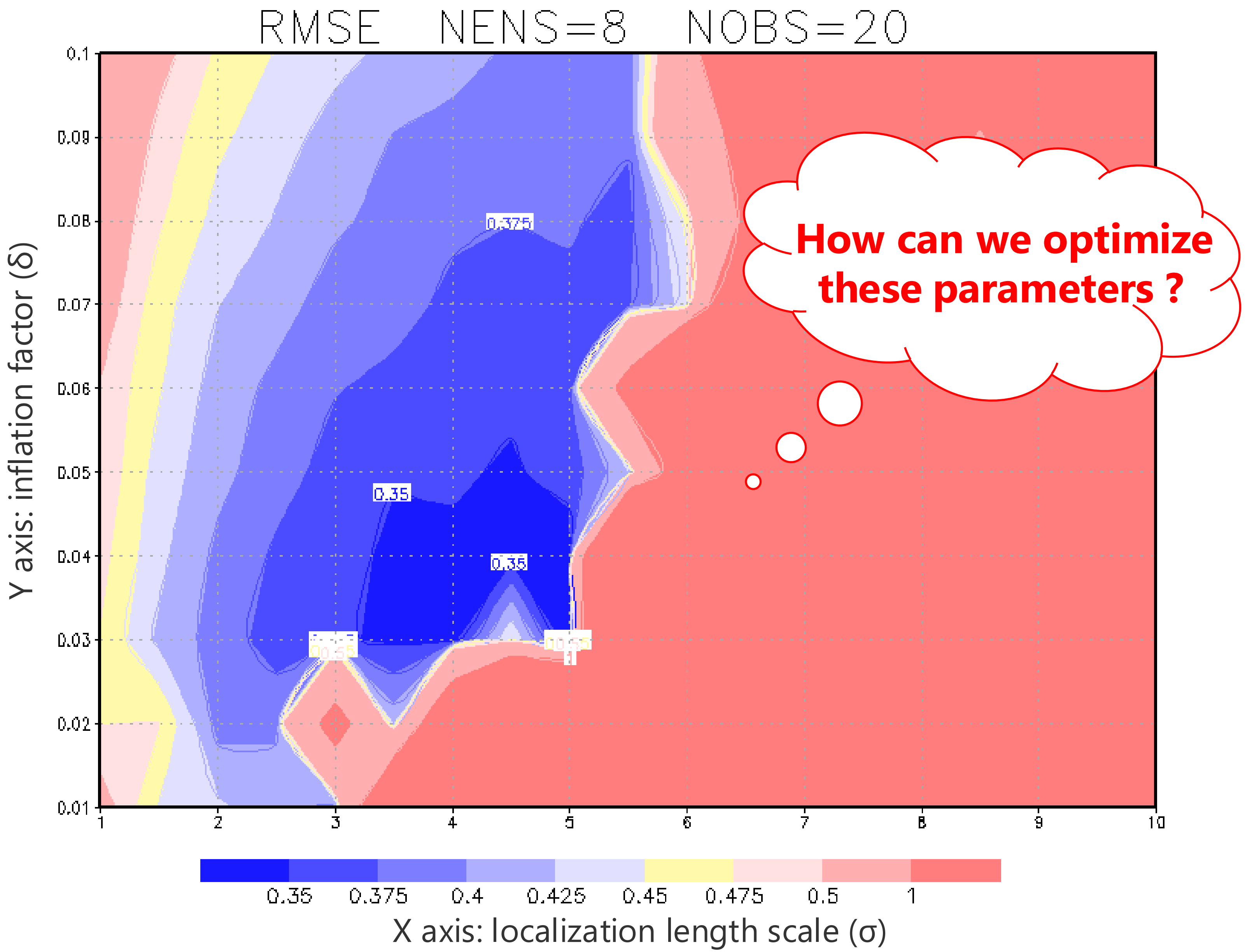
Today's Goal

- **Lecture: adaptive inflation**
  - to understand several inflation methods
  - to understand adaptive inflations

- **Training Course: Lorenz 96**
  - to implement adaptive inflation into L96

Review

Motivation (1)



Innovation statistics

$$\mathbf{d}^{o-b} = \mathbf{y}^{o} - \mathbf{H}\mathbf{x}^{b}$$

b: background

$$\mathbf{d}^{o-a} = \mathbf{y}^{o} - \mathbf{H}\mathbf{x}^{a}$$

a: analysis

$$\mathbf{d}^{a-b} = \mathbf{H}\mathbf{x}^{a} - \mathbf{H}\mathbf{x}^{b}$$

o: observation

Desroziers' innovation statistics (Desroziers et al. 2005)

$$\left\langle \mathbf{d}^{o-b}(\mathbf{d}^{o-b})^{T} \right\rangle = \mathbf{HBH}^{T} + \mathbf{R} \qquad \left\langle \mathbf{d}^{a-b}(\mathbf{d}^{o-b})^{T} \right\rangle \approx \mathbf{HBH}^{T}$$

$$\left\langle \mathbf{d}^{o-a}(\mathbf{d}^{o-b})^{T} \right\rangle \approx \mathbf{R} \qquad \left\langle \mathbf{d}^{a-b}(\mathbf{d}^{o-a})^{T} \right\rangle \approx \mathbf{HAH}^{T}$$

See A09 Innovation Statistics

Inflation Methods

Inflation Methods

Multiplicative Inflation

$$\mathbf{P}_{t,inf}^{b} = \Delta\mathbf{P}_{t}^{b}$$

Relaxation to Prior Perturbation (RTPP)

$$\delta\mathbf{X}_{t,inf}^{a} = (1-\alpha)\delta\mathbf{X}_{t}^{a} + \alpha\delta\mathbf{X}_{t}^{b}$$

Relaxation to Prior Spread (RTPS)

$$\delta\mathbf{X}_{t,inf,i}^{a} = \frac{\alpha\sigma_i^b + (1-\alpha)\sigma_i^a}{\sigma_i^a}\delta\mathbf{X}_{t,i}^{a}$$

Additive Inflation

$$\delta\mathbf{X}_{t,inf}^{a} = \delta\mathbf{X}_{t}^{a} + \delta\mathbf{X}^{rand}$$

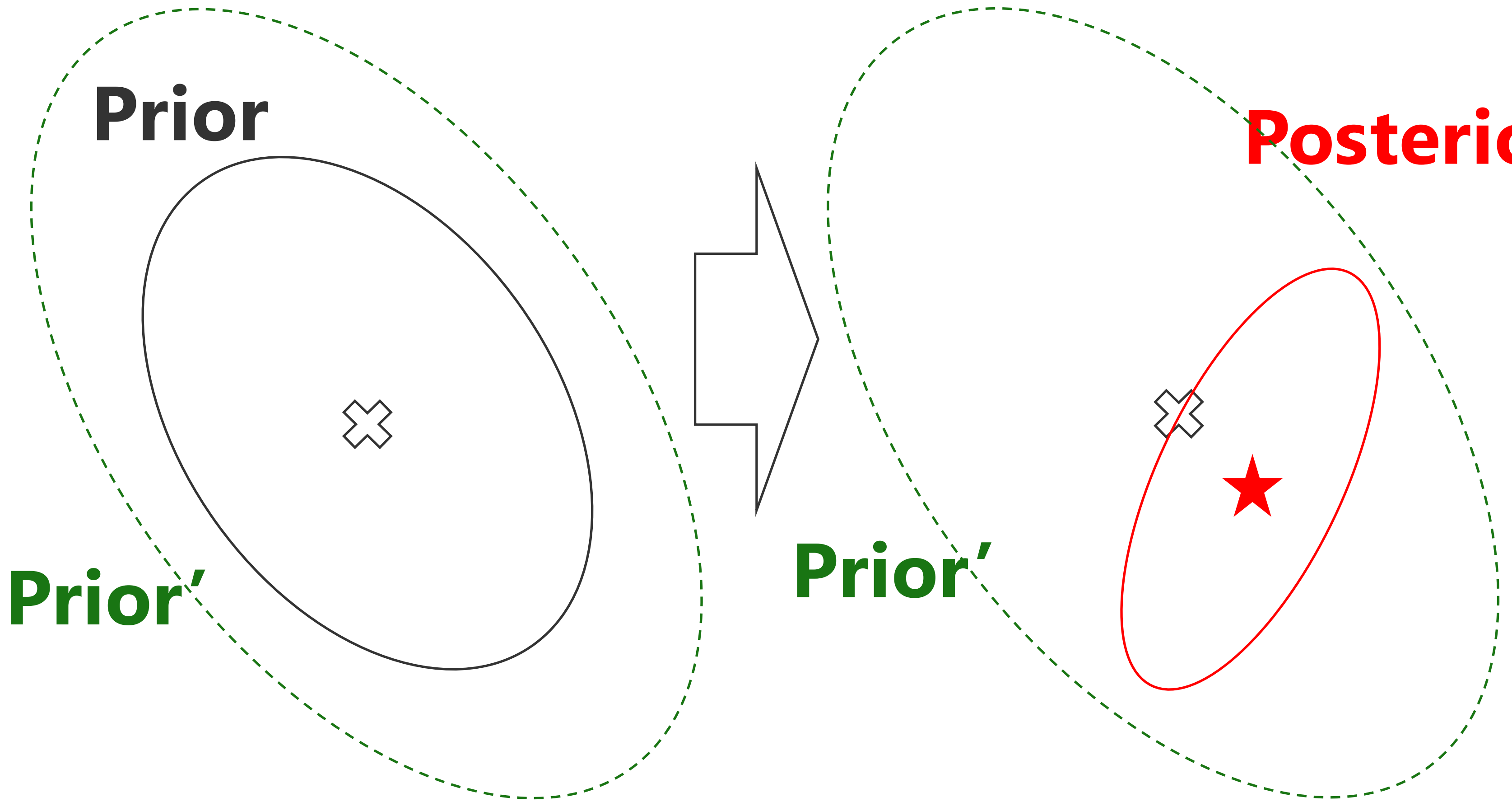
Multiplicative Inflation

Anderson and Anderson (1999; MWR)

Multiplicative Inflation

$$\mathbf{P}_{t,inf}^{b} = \Delta\mathbf{P}_{t}^{b}$$

$$\xrightarrow{\text{FCST}} \mathbf{P}_t^b \xrightarrow{\text{inflation}} \mathbf{P}_{t,inf}^b \xrightarrow{\text{DA}} \mathbf{P}_t^a \qquad \xrightarrow{\text{FCST}} \mathbf{P}_t^b$$



RTPP

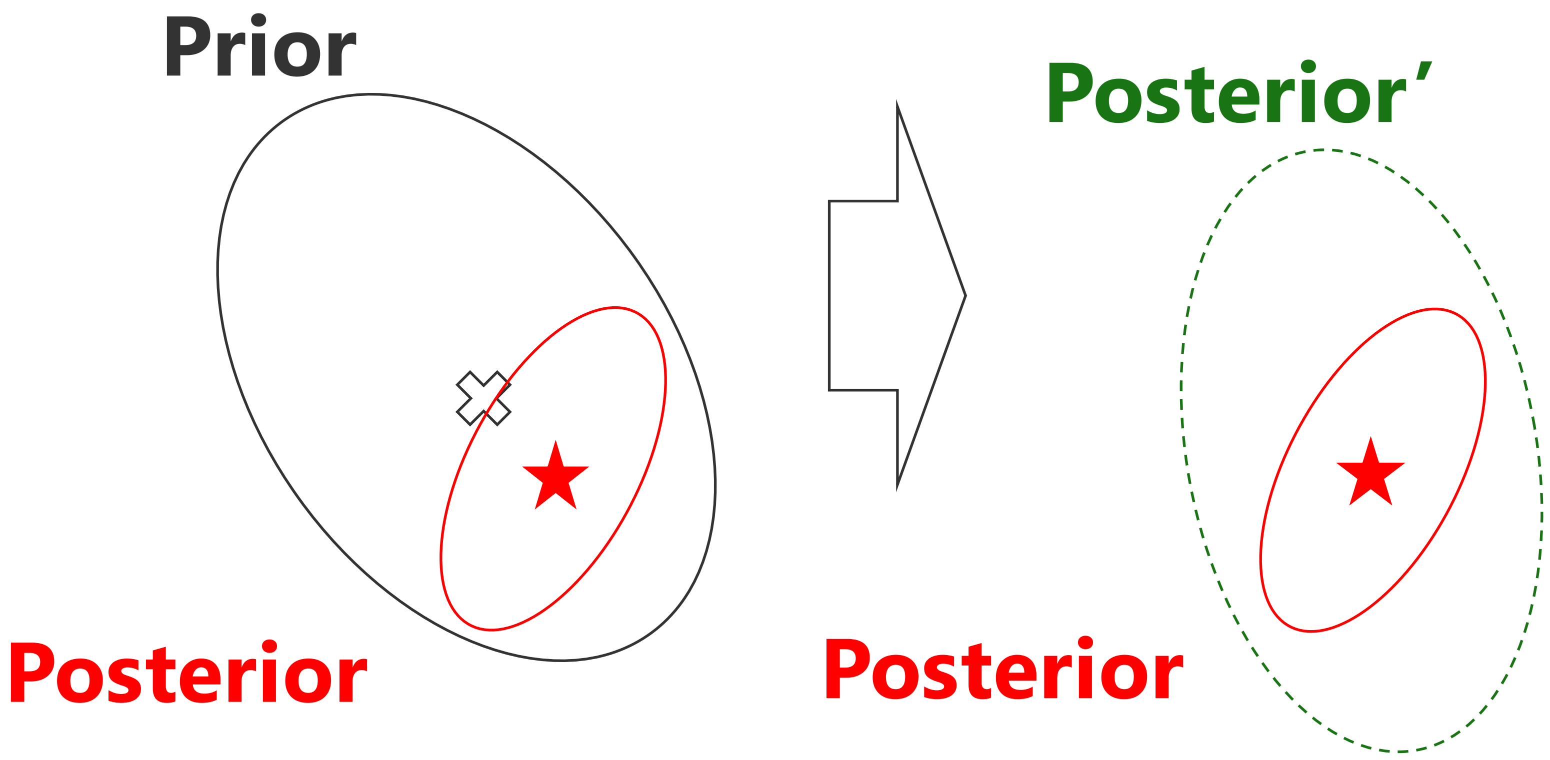
Zhang et al. (2004; MWR)

Relaxation to Prior Perturbation (RTPP)

$$\delta\mathbf{X}_{t,inf}^{a} = (1-\alpha)\delta\mathbf{X}_{t}^{a} + \alpha\delta\mathbf{X}_{t}^{b}$$

$$\xrightarrow{\text{FCST}} \mathbf{P}_t^b$$

$$\xrightarrow{\text{DA}} \mathbf{P}_t^a \xrightarrow{\text{relaxation}} \mathbf{P}_{t,inf}^a \xrightarrow{\text{FCST}} \mathbf{P}_t^b$$



RTPS

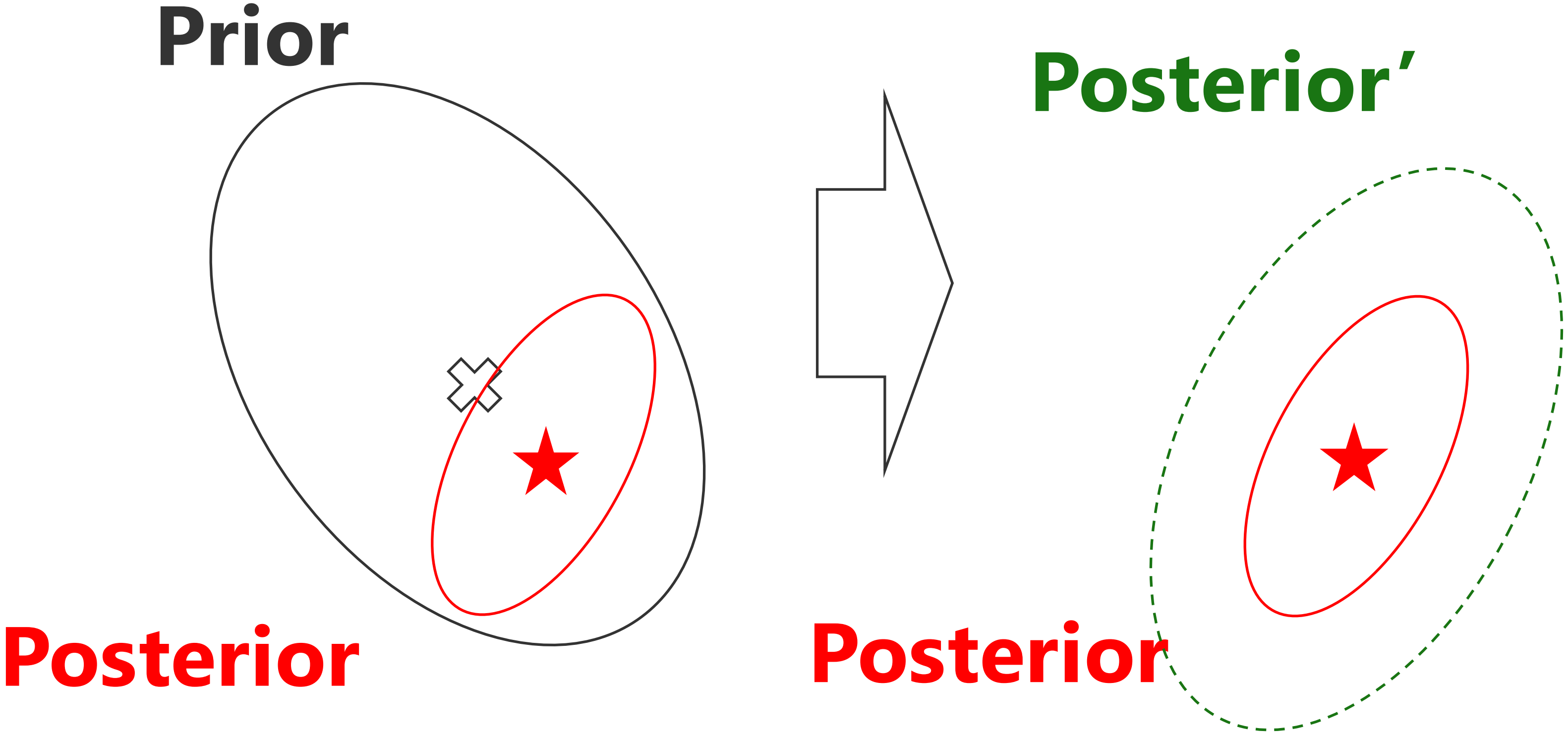
Whitaker and Hamill (2012; MWR)

Relaxation to Prior Spread (RTPS)

$$\delta \mathbf{X}^{a}_{t,inf,i} = \frac{\alpha\sigma_i^b + (1-\alpha)\sigma_i^a}{\sigma_i^a}\delta \mathbf{X}^{a}_{t,i}$$

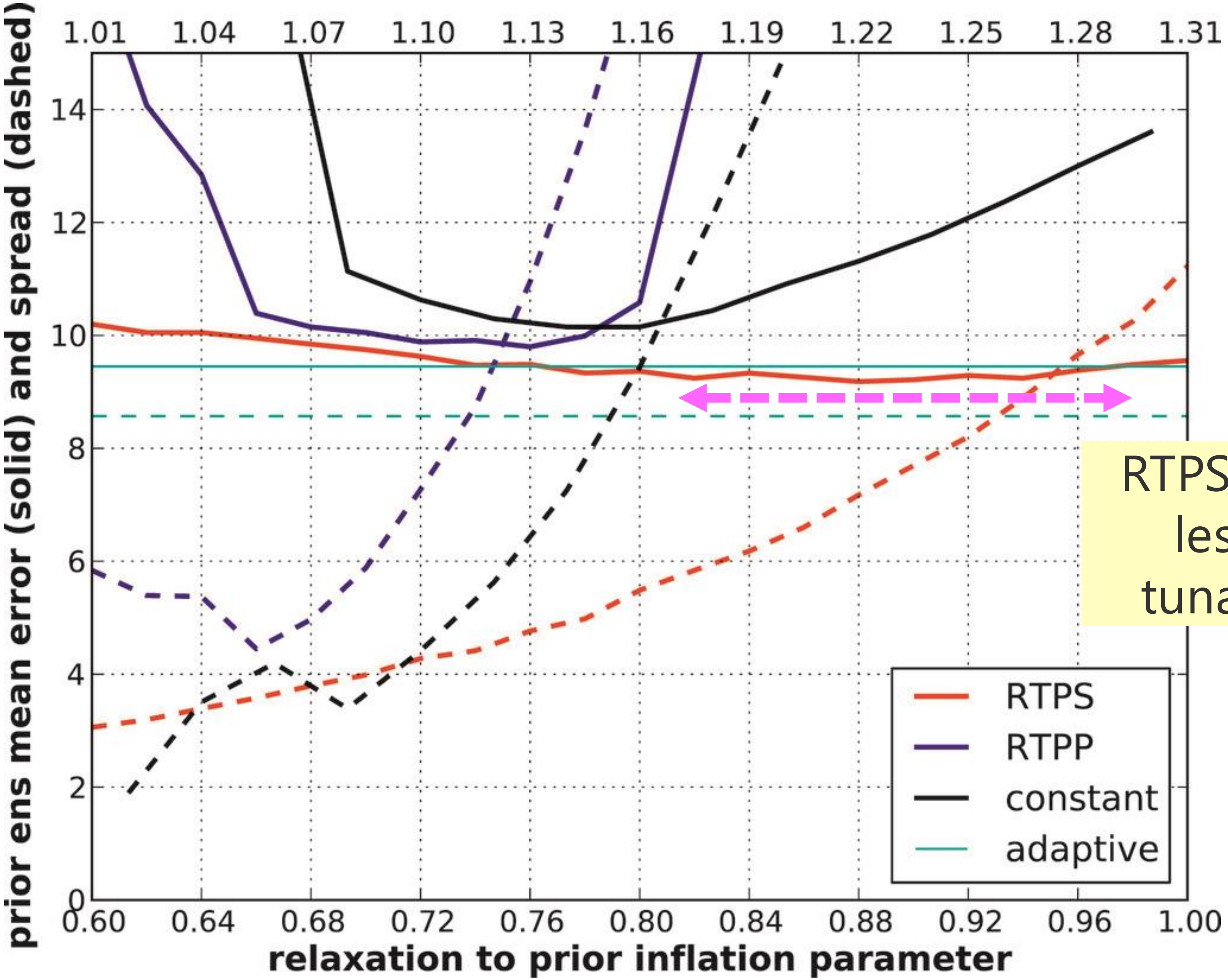
σ_i: ensemble spread of ith variable

$\xrightarrow{\text{FCST}} \mathbf{P}_t^b$ $\xrightarrow{\text{DA}} \mathbf{P}_t^a \xrightarrow{\text{relaxation}} \mathbf{P}_{t,inf}^a \xrightarrow{\text{FCST}} \mathbf{P}_t^b$



Sensitivity to tunable parameters

Whitaker and Hamill (2012; MWR)



RTPS is known to be less sensitive to tunable parameter

Adaptive Inflations

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EnKF

Ensemble Prediction (state)

$$\mathbf{x}_t^{b(i)} = M\left(\mathbf{x}_{t-1}^{a(i)}\right) \quad \text{for } i = 1,\ldots,m$$

Prediction of Error Covariance (implicitly)

$$\mathbf{P}_t^b \approx \mathbf{Z}_t^b\left(\mathbf{Z}_t^b\right)^T$$

adaptive multiplicative inflations

Anderson (2007, 2009)
Li et al. (2009)
Miyoshi (2011)

Kalman Gain

$$\mathbf{K}_t = \mathbf{Z}_t^b(\mathbf{Y}_t^b)^T\left[\mathbf{Y}_t^b(\mathbf{Y}_t^b)^T + \mathbf{R}\right]^{-1}$$

$$= \mathbf{Z}_t^b\left[\mathbf{I} + (\mathbf{Y}_t^b)^T\mathbf{R}^{-1}\mathbf{Y}_t^b\right]^{-1}(\mathbf{Y}_t^b)^T\mathbf{R}^{-1}$$

Analysis (state)

$$\mathbf{x}_t^a = \mathbf{x}_t^b + \mathbf{K}_t(\mathbf{y}_t^o - H(\mathbf{x}_t^b))$$

Analysis Error Covariance

$$\mathbf{P}_t^a = [\mathbf{I} - \mathbf{K}_t\mathbf{H}]\mathbf{P}_t^b$$

Adaptive RTPP & RTPS

Ying and Zhang (2015), Kotsuki et al. (2016)

Miyoshi (2011)'s adaptive multiplicative inflation

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$\Delta_t^a \rightarrow \Delta_{t+1}^b$

for the next time step

Data assimilation at *t*

using Δ_t^b so that $\mathbf{P}_{t,inf}^b = \Delta_t^b \mathbf{P}_t^b$

Estimation of the factor

$$\Delta_t^o = \frac{tr\left(\mathbf{d}^{o-b}\left(\mathbf{d}^{o-b}\right)^T \circ \mathbf{R}^{-1}\right) - p}{tr\left(\mathbf{HBH}^T \circ \mathbf{R}^{-1}\right)}$$

$$\left\langle \mathbf{d}^{o-b}\left(\mathbf{d}^{o-b}\right)^T \right\rangle = \mathbf{HBH}^T + \mathbf{R}$$

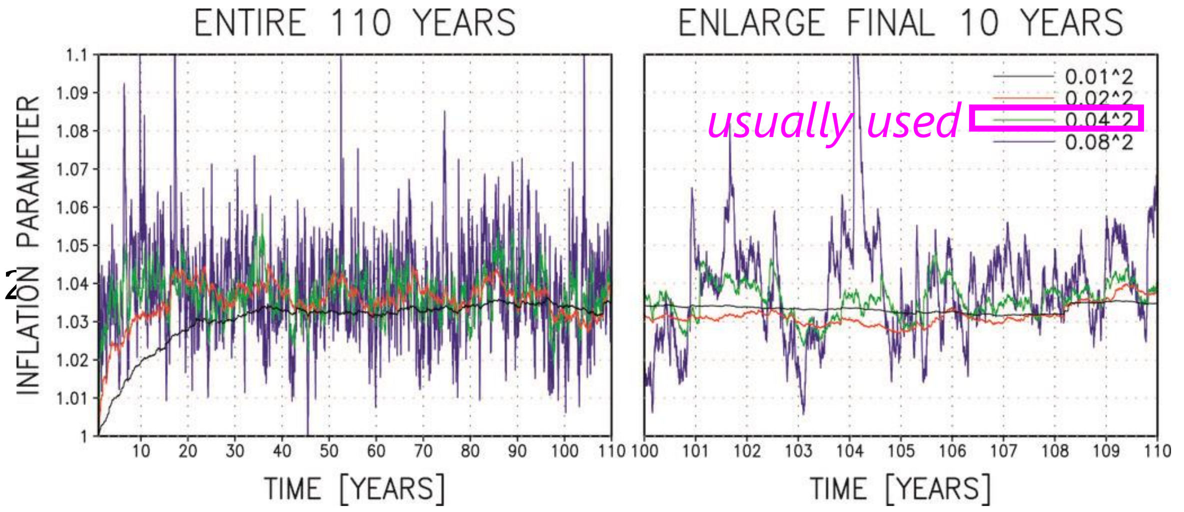
analysis of the inflation factor

$$\Delta_t^a = \frac{\Delta_t^b v_i^o + \Delta_t^o v_i^b}{v_i^o + v_i^b}$$

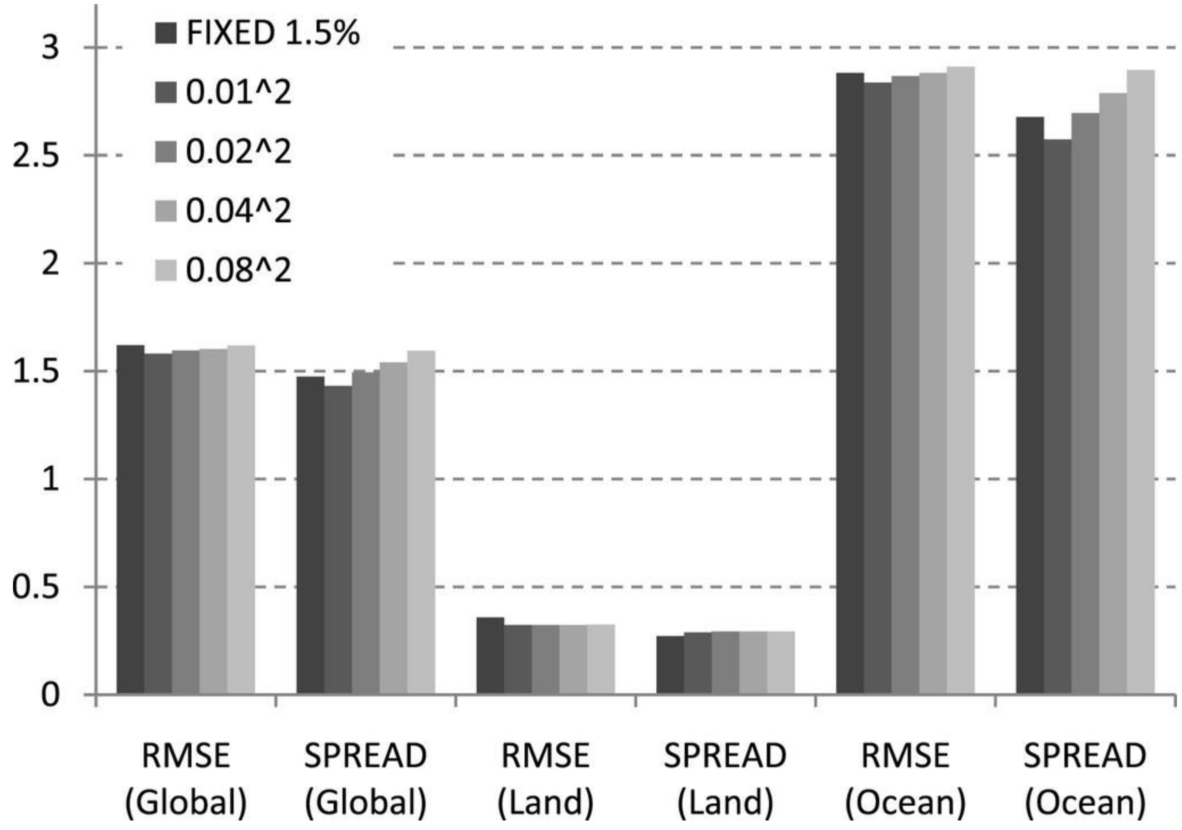
a tunable parameter

where $$v_t^o = \frac{2}{p}\left[\frac{\Delta_t^b tr\left(\mathbf{HBH}^T \circ \mathbf{R}^{-1}\right) + p}{tr\left(\mathbf{HBH}^T \circ \mathbf{R}^{-1}\right)}\right]^2$$

Miyoshi (2011) w/ L96



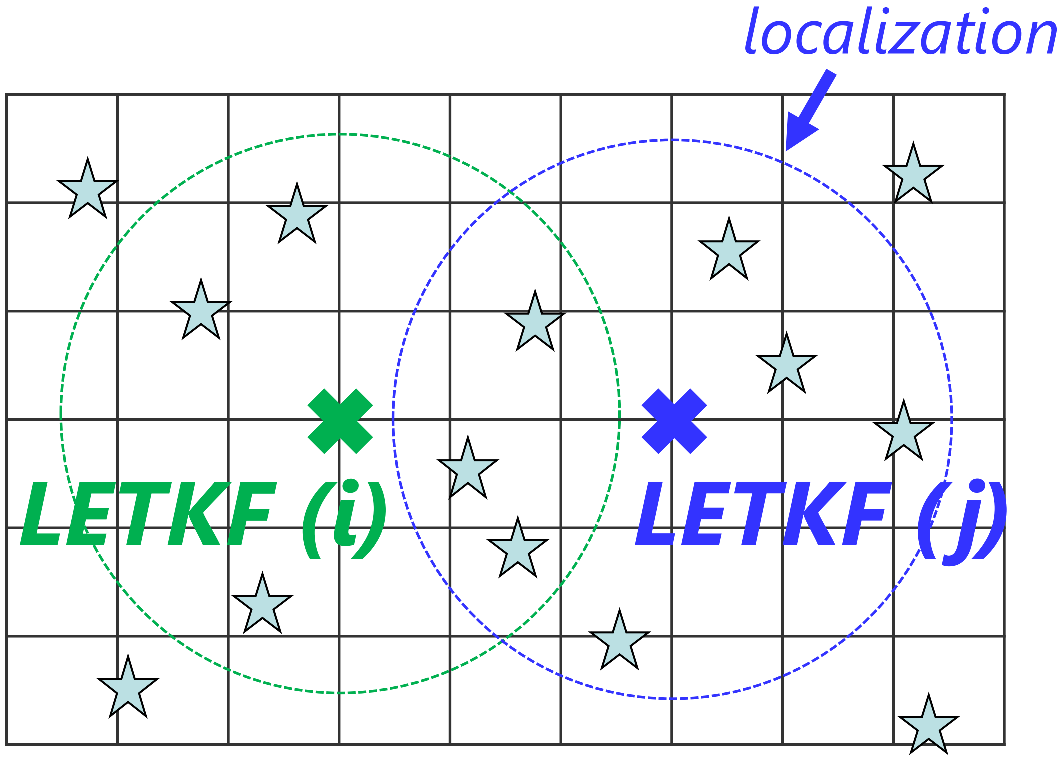
Lorenz 96 Experiments in Miyoshi (2011)



more inflations
over land

Prior inflation variance	Inflation over land (%)	Inflation over the ocean (%)
0.01^2	4.54	2.22
0.02^2	4.61	2.56
0.04^2	4.57	2.88
0.08^2	4.56	3.17

Tips for implementations



for adaptive multiplicative inflations,
OMB-OMB statistics is computed using
obs in localization radius
to represent spatially-varying inflation factors

Miyoshi (2011) used

$$\left\langle \mathbf{d}^{o-b}\left(\mathbf{d}^{o-b}\right)^T \right\rangle = \mathbf{HBH}^T + \mathbf{R}$$

JMA used (personal communication)

$$\left\langle \mathbf{d}^{a-b}\left(\mathbf{d}^{o-b}\right)^T \right\rangle \approx \mathbf{HBH}^T$$

this approach is more robust for realistic exp.
in which larger obs error is used for stabilization

where analysis in obs space is computed by

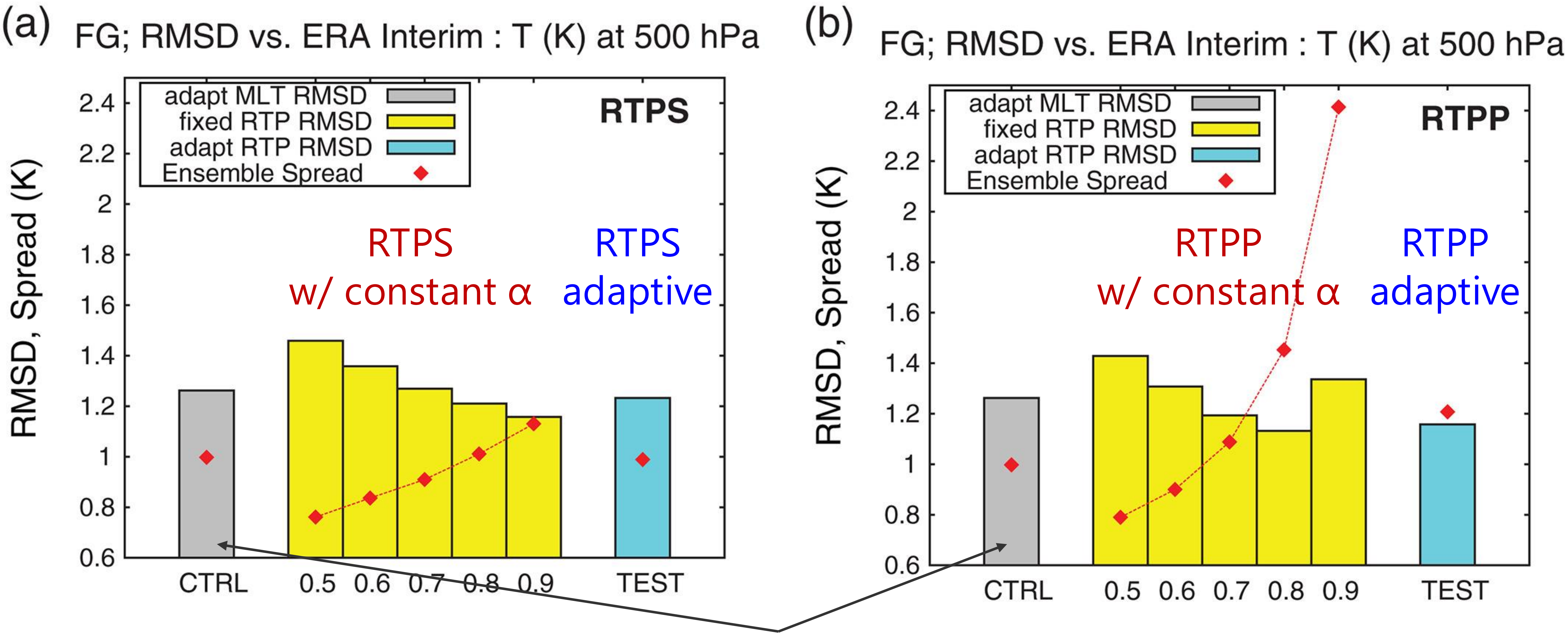
$$\delta\bar{\mathbf{x}}^a = \mathbf{Z}^b\widetilde{\mathbf{w}} \longrightarrow \mathbf{H}\delta\bar{\mathbf{x}}^a = \mathbf{H}\mathbf{Z}^b\widetilde{\mathbf{w}}$$

Adopted from
Kotsuki et al. (2020; QJRMS)

Comparisons w/ an NWP system
- NICAM-LETKF

A case with NICAM-LETKF

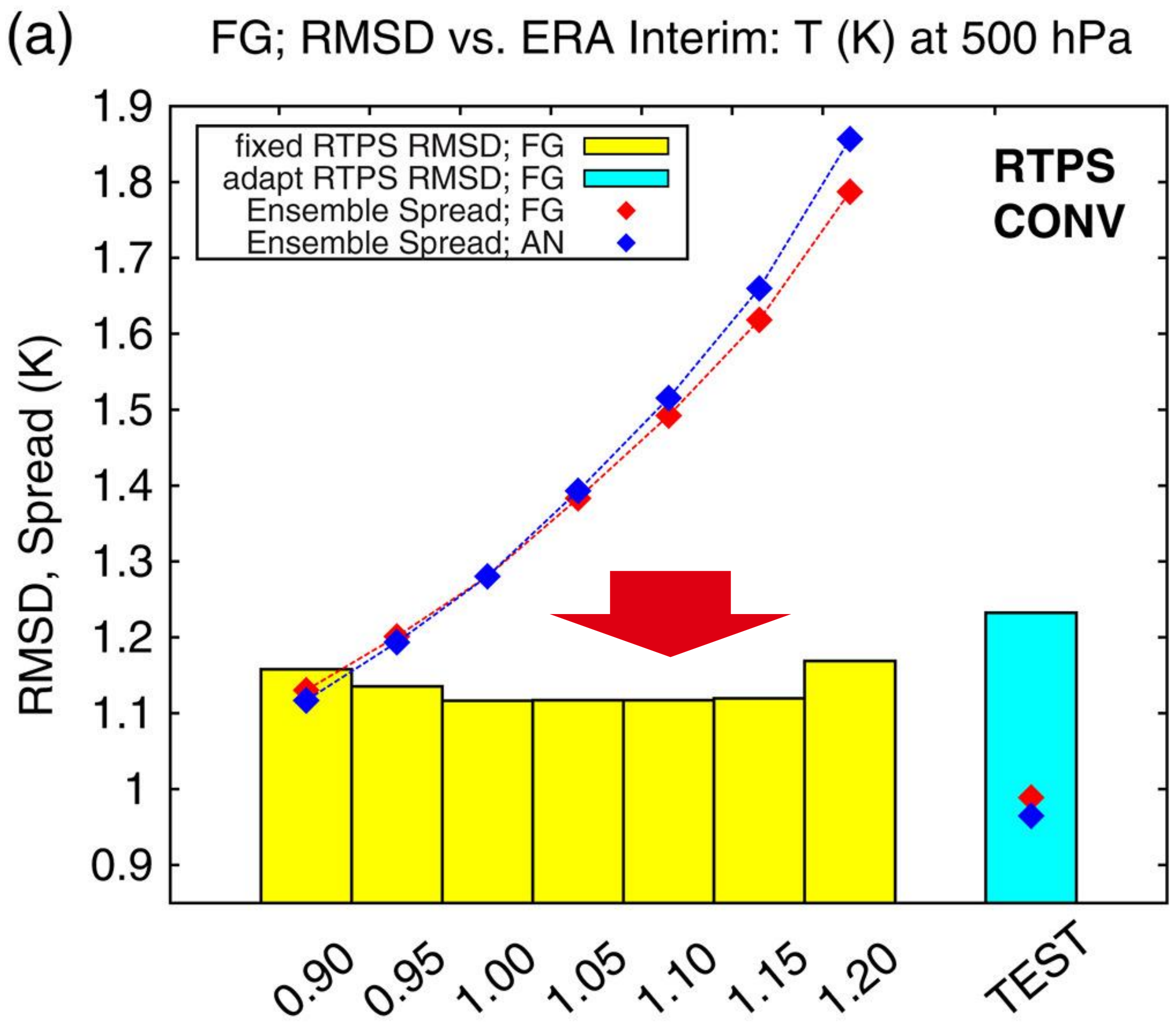
Kotsuki et al. (2017; QJRMS)

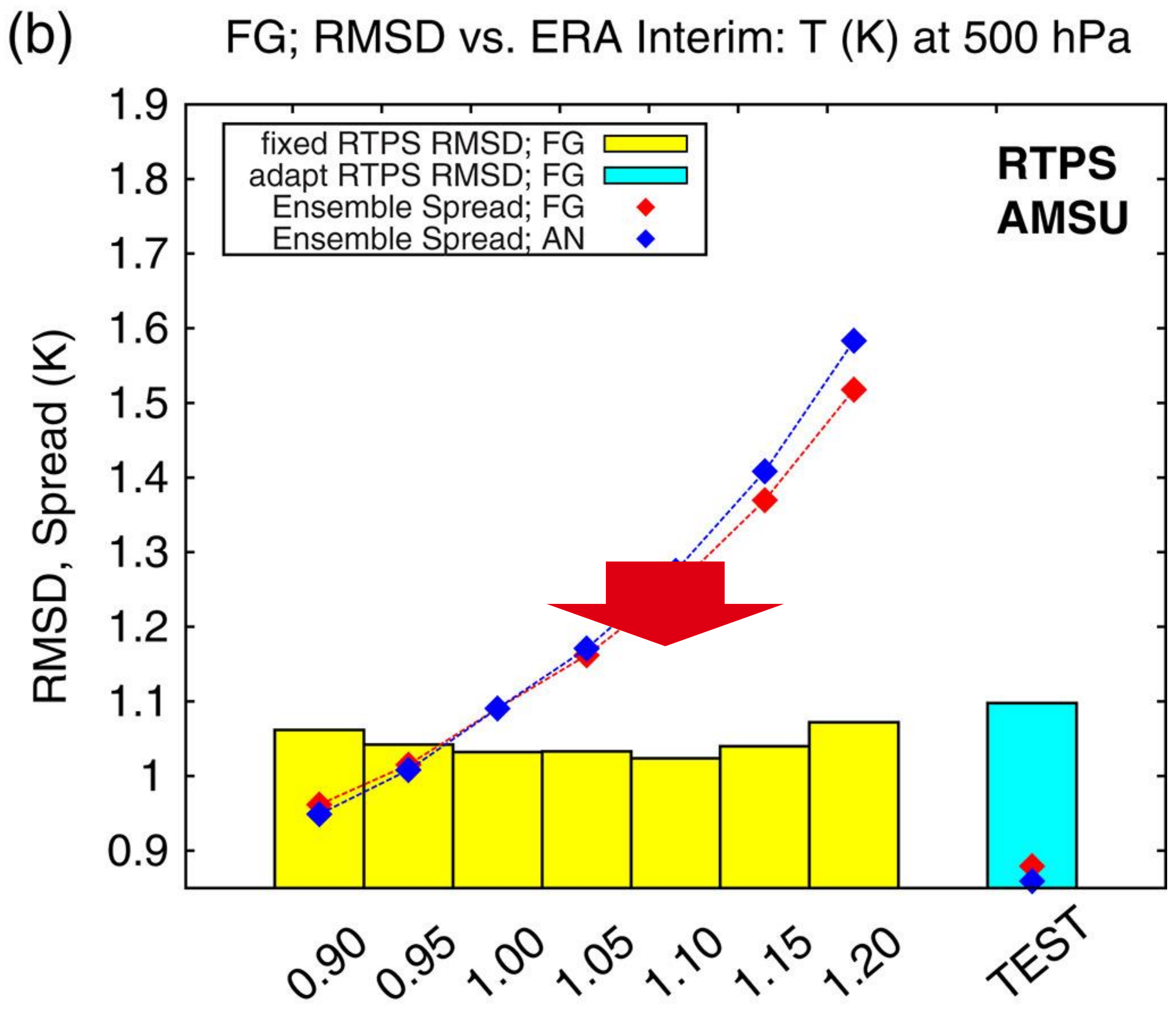


adaptive multiplicative
(Miyoshi 2011)

A case with NICAM-LETKF

Kotsuki et al. (2017; QJRMS)





The optimal relaxation parameter can be $\alpha > 1.00$,
meaning that posterior spread is than prior spread

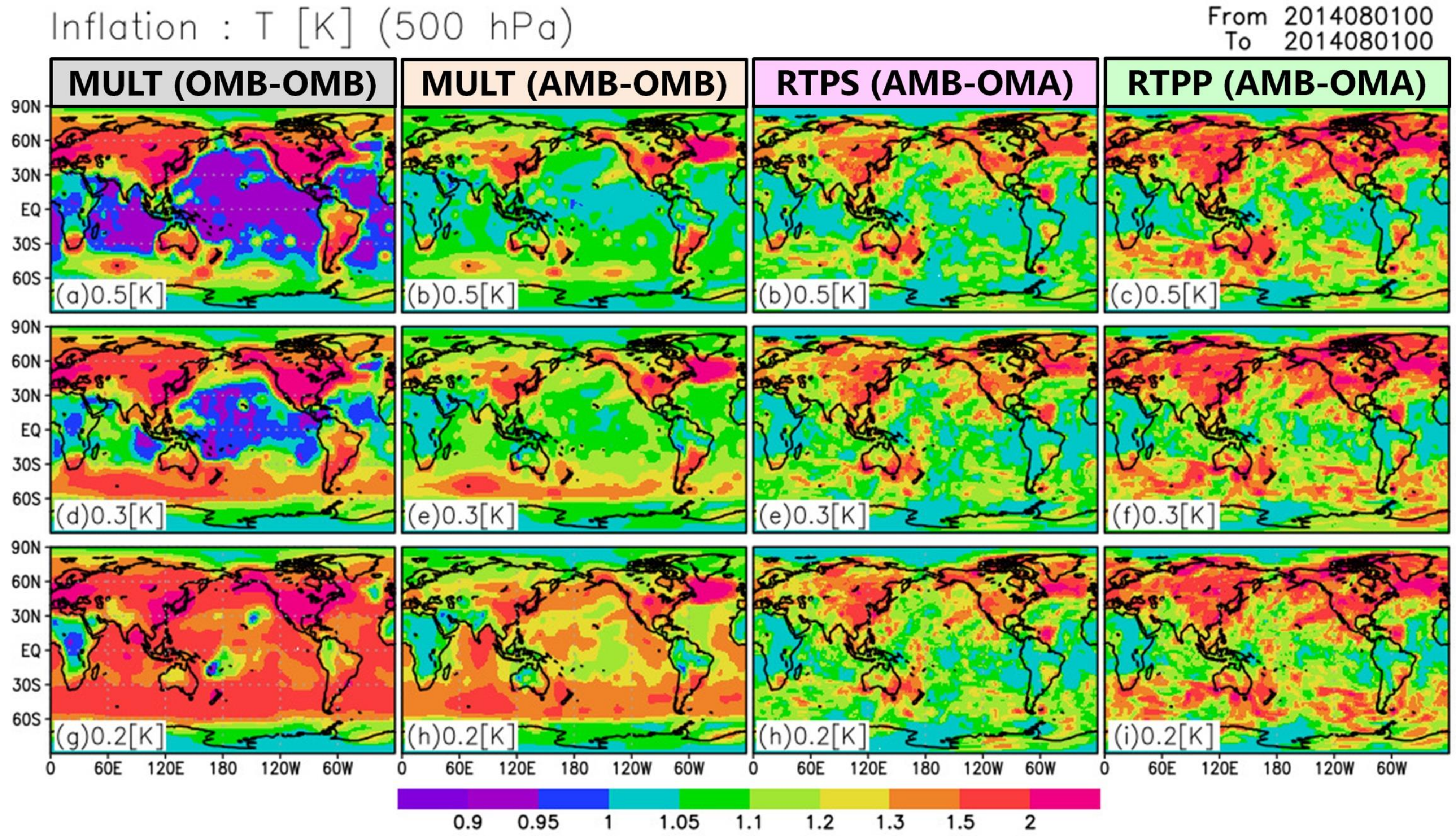
Comparison of inflation factors

Kotsuki et al. (2017; QJRMS)

AMSU-A SD=0.5K

AMSU-A SD=0.3K

AMSU-A SD=0.2K

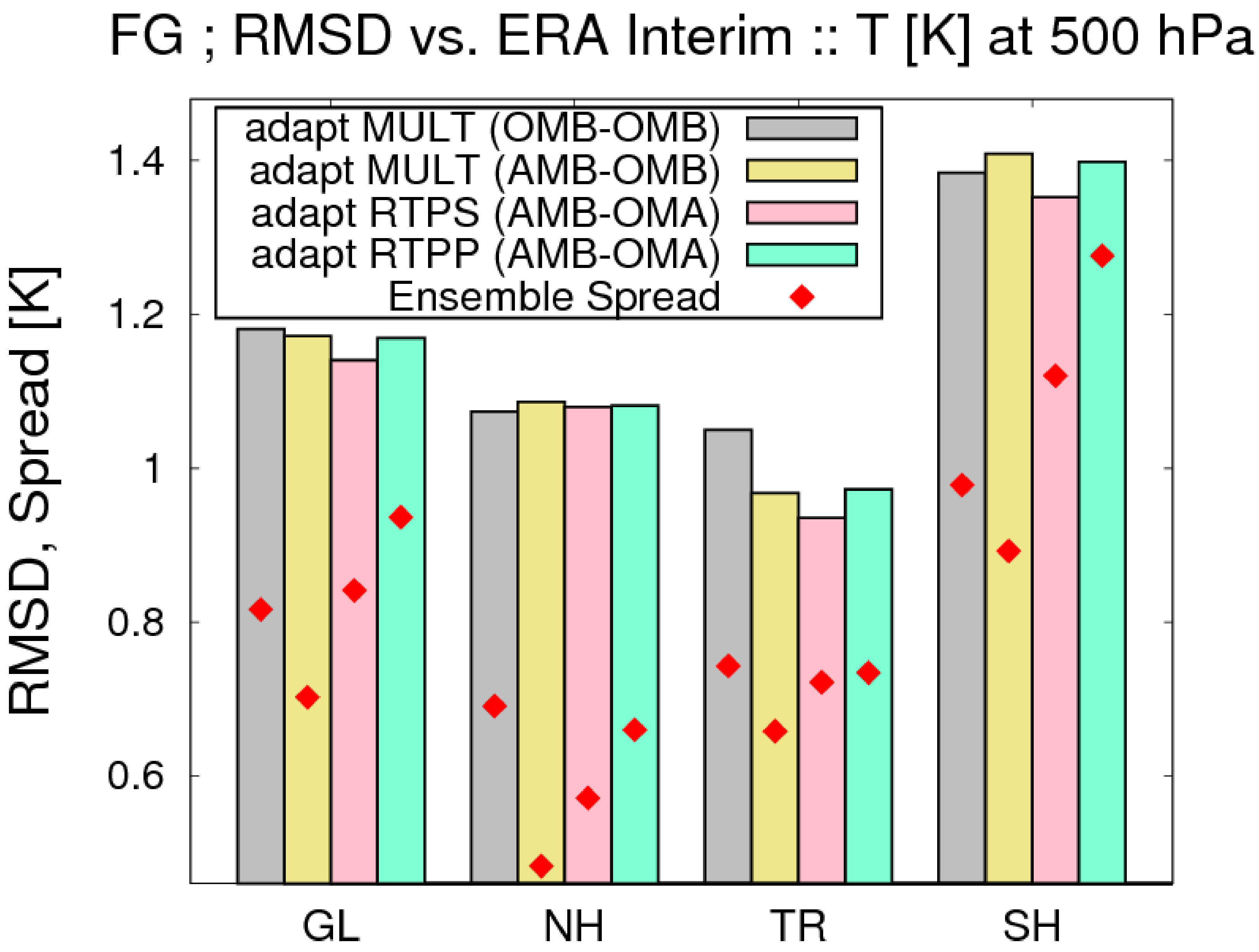


very sensitive to SD☹

less sensitive to SD☺

RMSD vs ERA Interim

Kotsuki et al. (2017; QJRMS)



Interpretation by Innovation Statistics

See A09 Innovation Statistics

Innovation Statistics

$$\left\langle \mathbf{d}^{o-b}(\mathbf{d}^{o-b})^T \right\rangle = \mathbf{HBH}^T + \mathbf{R}$$

overestimated when $\tilde{\mathbf{R}}$ is large

$$\left\langle \mathbf{d}^{o-a}(\mathbf{d}^{o-b})^T \right\rangle = \tilde{\mathbf{R}}\left(\mathbf{H}\tilde{\mathbf{B}}\mathbf{H}^T + \tilde{\mathbf{R}}\right)^{-1}(\mathbf{HBH}^T + \mathbf{R}) = \mathbf{R}^e$$

understimated when $\tilde{\mathbf{R}}$ is large

$$\left\langle \mathbf{d}^{a-b}(\mathbf{d}^{o-b})^T \right\rangle = \mathbf{H}\tilde{\mathbf{B}}\mathbf{H}^T\left(\mathbf{H}\tilde{\mathbf{B}}\mathbf{H}^T + \tilde{\mathbf{R}}\right)^{-1}(\mathbf{HBH}^T + \mathbf{R}) = \mathbf{HB}^e\mathbf{H}^T$$

canceled out!

$$\left\langle \mathbf{d}^{a-b}(\mathbf{d}^{o-a})^T \right\rangle = \mathbf{HB}^e\mathbf{H}^T(\mathbf{HBH}^T + \mathbf{R})^{-1}\mathbf{R}^e = \mathbf{HA}^e\mathbf{H}^T$$

Their Relationship

$$\mathbf{R}^e + \mathbf{HB}^e\mathbf{H}^T = \mathbf{R} + \mathbf{HBH}^T$$

Balanced!

$\mathbf{R}, \mathbf{B}, \mathbf{A}$: Truth
$\tilde{\mathbf{R}}, \tilde{\mathbf{B}}, \tilde{\mathbf{A}}$: DA (maybe imperfect)
$\mathbf{R}^e, \mathbf{B}^e, \mathbf{A}^e$: Estimated

o, R: obs
b, B: background
a, A: analysis

For further studies

Why does EnKF suffer from analysis overconfidence?

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Quarterly Journal of the Royal Meteorological Society

RMetS Royal Meteorological Society

RESEARCH ARTICLE | Full Access

Why does EnKF suffer from analysis overconfidence? An insight into exploiting the ever-increasing volume of observations

Daisuke Hotta, Yoichiro Ota

First published: 27 December 2020 | https://doi.org/10.1002/qj.3970 | Citations: 9

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Thank you for your attention!

Presented by Shunji Kotsuki
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Further information is available at
https://kotsuki-lab.com/

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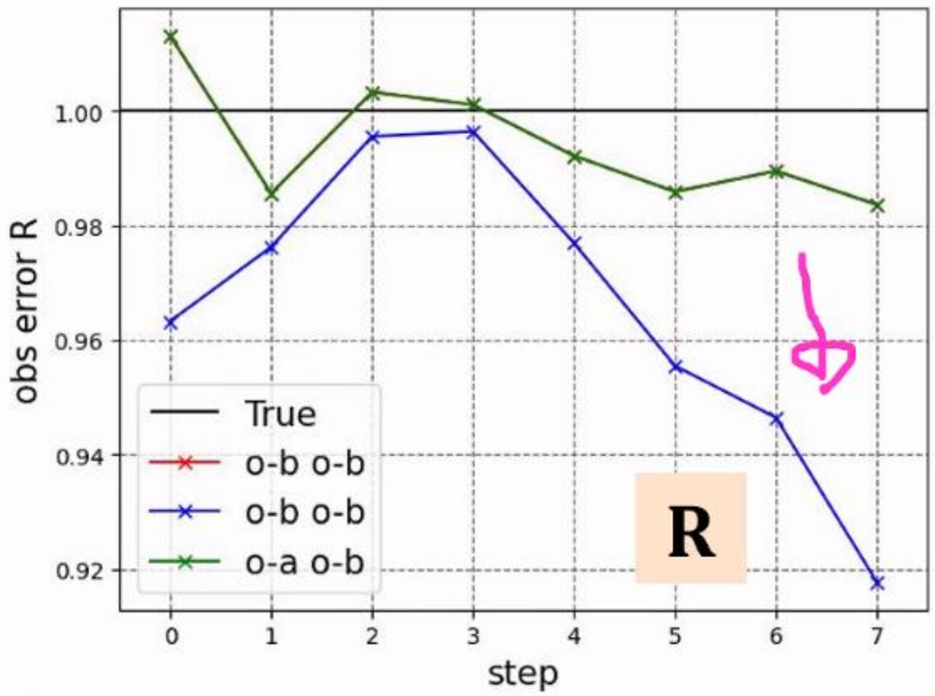
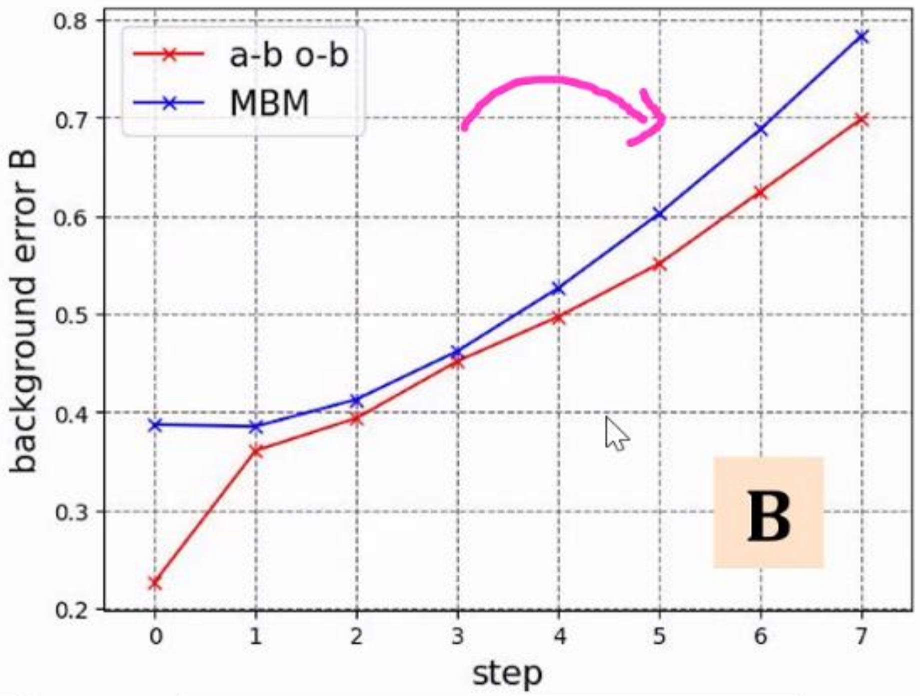
Appendix





red: $\left\langle \mathbf{d}^{o-b}\left(\mathbf{d}^{o-b}\right)^T \right\rangle - \left\langle \mathbf{d}^{a-b}\left(\mathbf{d}^{o-b}\right)^T \right\rangle$ から推定

blue: $\left\langle \mathbf{d}^{o-b}\left(\mathbf{d}^{o-b}\right)^T \right\rangle - \mathbf{HMBM}^T\mathbf{H}^T$ から推定

green: $\left\langle \mathbf{d}^{o-a}\left(\mathbf{d}^{o-b}\right)^T \right\rangle$ から推定

red: $\left\langle \mathbf{d}^{a-b}\left(\mathbf{d}^{o-b}\right)^T \right\rangle$ から推定

blue: $\mathbf{HMBM}^T\mathbf{H}^T$ から推定

$window = 8\ step$

$B_{ii} = 0.15$

$$\left\langle \mathbf{d}^{o-b}\left(\mathbf{d}^{o-b}\right)^T \right\rangle = \mathbf{R} + \mathbf{HBH}^T \quad \left\langle \mathbf{d}^{a-b}\left(\mathbf{d}^{o-b}\right)^T \right\rangle = \mathbf{HBH}^T \quad \left\langle \mathbf{d}^{o-a}\left(\mathbf{d}^{o-b}\right)^T \right\rangle = \mathbf{R}$$